LESSON 3.5

Compare Ratio Relationships

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can compare ratios by finding unit rates or using equivalent ratios.

WHY THIS MATTERS Two friends used the paint shown to create shades of purple paint. Whose mixture contains the most blue paint?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Unit rate *Spanish: Tasa unitaria*	A rate for just 1 of something, like cost for 1 item.	$3 per 1 pound → unit rate is $3/lb
Equivalent ratio *Spanish: Razón equivalente*	Two ratios that mean the same thing.	4:6 and 2:3 are equivalent (both simplify to 2:3)
Compare *Spanish: Comparar*	To look at ratios and see which is bigger, smaller, or equal.	3:4 vs. 2:5 — which has more of the first ingredient per unit?
Simplify *Spanish: Simplificar*	To write a ratio with smaller numbers that still compares the same way. 6:8 and 3:4 are the same ratio.	8:12 → divide both by 4 → 2:3
Common denominator *Spanish: Denominador común*	A bottom number that two fractions can share so you can compare them.	To compare 3/5 and 4/7, use LCD 35: 21/35 vs 20/35
Common multiple *Spanish: Múltiplo común*	A number that both terms can be scaled to.	2 and 3 both reach 6

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Mix A has 2 soap to 5 water. Mix B has 3 soap to 7 water. Which is stronger?

1. Common multiple of 2 and 3: 6.
2. Mix A × 3 = 6 : 15. Mix B × 2 = 6 : 14.
3. Compare water: 14 < 15.

Answer: Mix B is stronger (less water for 6 soap)

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Store A sells 2 pens for $6. Store B sells 3 pens for $12. Which is the better deal?

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- I scaled both ratios to _____ _____ .
- Mix _____ has _____ and Mix _____ has _____ .
- _____ is stronger because _____ .
- You cannot compare them directly because _____ .

Word bank common multiple | least common multiple | scale | match | compare | stronger | weaker | less | more | same | for every | fair

WATCH OUT!

A common mistake

A common mistake in Compare Ratios is thinking the ratio with the bigger raw number automatically wins

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Compare Ratio Relationships

Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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1 **Circle the letter of the best answer. Show how you know.**

Chef A uses 2 cups of cheese for every 8 crackers. Chef B uses 3 cups of cheese for every 9 crackers. Who uses more cheese per cracker?

(A) Chef A
(B) Chef B
(C) They use the same amount
(D) Cannot determine

STEPS TO TRY
1. You can't compare 2 cups and 3 cups directly — the chefs use different numbers of crackers.
2. Find each chef's cheese for ONE cracker: divide cheese by crackers for each, then compare the two rates.
3. Chef A: divide 2 ÷ 8. Chef B: divide 3 ÷ 9. Compare those two decimals to see who uses more per cracker.

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Which ratio is greater: 3:4 or 5:8?

(A) 5:8
(B) 3:4
(C) They are equal
(D) Cannot determine

STEPS TO TRY

1. The second numbers (4 and 8) don't match, so the ratios aren't ready to compare yet.
2. Rewrite the ratios with the same second number — 8 works nicely, since 4 × 2 = 8 — then compare the first numbers.
3. Scale 3:4 by 2: multiply both parts to get an out-of-8 ratio. Then compare its first number with the 5 in 5:8.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Which lemonade recipe is more lemony? Recipe A: 2 lemons for 6 cups of water. Recipe B: 3 lemons for 7 cups of water.

(A) Recipe B
(B) Recipe A
(C) They are the same
(D) Cannot determine

STEPS TO TRY

1. Recipe B has more lemons, but it also has more water — so compare rates, not raw counts.
2. Find lemons per ONE cup of water for each recipe: divide lemons by cups. The bigger rate is more lemony.
3. Recipe A: divide 2 ÷ 6. Recipe B: divide 3 ÷ 7. Compare the two decimals you get.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Which is the better deal: 4 apples for $3 or 6 apples for $5?

(A) 6 apples for $5

(B) 4 apples for $3

(C) They cost the same

(D) Cannot determine

STEPS TO TRY

1. 'Better deal' means the LOWER cost for one apple — not simply more apples.
2. Find the price of one apple in each deal: divide dollars by apples, then pick the smaller price.
3. Deal 1: divide 3 ÷ 4 to get dollars per apple. Do the same with 5 ÷ 6, then compare the two prices.

SHOW YOUR WORK

EXPLAIN YOUR THINKING

The two recipes use different amounts, so you cannot compare them directly. How did building equivalent ratios let you compare them fairly?

SENTENCE STARTER I scaled both ratios to the same amount of ___, then compared the ___; Chef ___'s recipe is more chocolatey because ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help